

NETAJI SUBHAS UNIVERSITY OF TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

B.TECH. PROJECT REPORT



A 2.5D CNN–Transformer Framework with Attention and Progressive Pruning for Lung Nodule Detection

Submitted by:	Under the supervision of:
Vishal Kumar - 2023UCA1854 Ryldham Bansal - 2023UCS1599 Kushagra Sabharwal - 2023UCA1851	Dr. Surendra Nagar Assistant Professor

CONTENTS

1.	Introduction	3
2.	Motivation	4
2.1.	Limited Context in 2D Processing	4
2.2.	Local and Contextual Feature Modeling	4
2.3.	Model Complexity and Efficiency	4
3.	Literature Survey	4
3.1.	CNN–Transformer and Attention-Based Classification Models	4
3.2.	Lightweight and Memory-Efficient Architectures	5
3.3.	Detection-Oriented (YOLO-Family) Architectures	6
3.4.	A Review of Detection Architectures	6
4.	Problem Statement.....	7
5.	Objective and Methodology.....	7
5.1.	Research Objectives.....	7
5.2.	Proposed Architecture	8
5.3.	Proposed Training Objective	9
5.4.	Progressive Pruning.....	10
5.5.	Evaluation Metrics	11
6.	Simulation Platform and Requirements	11
6.1.	Dataset.....	11
6.2.	Hardware Environment	11
6.3.	Software Environment	12
7.	Conclusion	12
8.	References	13

1. Introduction

Lung cancer still accounts for one of the major reasons for cancer deaths and detection of pulmonary nodules using low dose Computed Tomography (CT) screening has been proven to significantly contribute towards improving patient health status. Given the large number of two dimensional slices in a single scan of the thorax using CT, manual radiological analysis of all the slices in search of small and diverse in shape pulmonary nodules poses a time-consuming process with a certain degree of interobserver variability. This issue has stimulated nearly three decades of work on Computer Aided Detection (CAD) algorithms as well as recent developments in automated detection using deep learning.

The LUNA16 (LUng Nodule Analysis 2016) [9], which is a subset of the larger LIDC-IDRI database, became a standard benchmark for automated pulmonary nodule detection systems. The data set contains 888 CT scans with annotations of pulmonary nodules along with an official evaluation protocol based on the FROC curve and Competition Performance Metric (CPM).

A major concern pertaining to the architecture in this case is the manner in which volumetric CT scans are represented for a learning detector. In case of 2D representation, the computation takes place based on each slice without taking into account any context and is thus computationally efficient, but loses the context information present in the upper and lower slices relative to the candidate nodule. However, 3D representations preserve the volumetric context but lead to increased memory and computational demands, which becomes a huge constraint while working on standard research computing hardware. The compromise between the two is the use of 2.5D representation, where a few slices of the CT scan ($z-1$, z , $z+1$) stack result in multiple input channels.

In this study, a CNN-Transformer design is going to be developed for lung nodule detection using the LUNA16 dataset. Local spatial features would be extracted using a convolutional neural network backbone while Transformer would capture long-range contextual relationships between local spatial features. In addition, the study applies an attention mechanism which will help to emphasize the important features in relation to lung nodules as well as pruning for exploring the compression effects of the model. The overall goal of this study is not to make the assumption that the Transformer, attention, or pruning techniques will add value. It is actually to find out through ablation studies the role of each component.

The objective of this report is to discuss the reasons for the motivation for the research project, the literature background, the problem statement, the methodology and simulation requirements. This report is a reflection of the current state of an on-going research project within academia and holds no claim to be clinically relevant or conclusive in nature; all models presented in this paper are subject to experimentation.

2. Motivation

The motivation for this project arises from a combination of clinical relevance, well-documented technical challenges in pulmonary nodule detection, and a specific research gap identified in the recent literature.

2.1 Limited Context in 2D Processing

Nodules of the lungs are three-dimensional entities; however, information obtained in one CT slice is not much about the environment around the nodule. In 2.5D strategy, information from neighboring slices can be employed without requiring powerful computing as required by 3D networks.

2.2 Local and Contextual Feature Modeling

CNNs function very well when it comes to the extraction of local features like edges, textures, shapes, and anatomical structures as well. The detection of nodules also has the potential of making use of wider feature relationships. Therefore, this research seeks to experiment with the CNN-Transformer model architecture, in which:

- CNN → extracts local spatial features.
- Transformer → captures broader feature relations.

2.3 Model Complexity and Efficiency

The models that are required for imaging purposes can require considerable amounts of computing capacity. Therefore, apart from performance, other criteria can be considered too, including number of parameters, memory usage, computational expense, and latency. The objective of the current research project is to apply progressive pruning to reduce the redundancy in the model and check whether the reduction of parameters results in efficiency. However, the reduction of parameters does not guarantee faster inference. Therefore, both aspects will be evaluated in this project.

3. Literature Survey

3.1 CNN–Transformer and Attention-Based Classification Models

3.1.1 Attention-Enhanced InceptionNeXt Hybrid Model [1]

Ozdemir et al. [1] present a novel deep learning approach for detecting lung cancer using a fusion of InceptionNeXt style multi-scale convolutional layers with Vision Transformer style grid attention and block attention, based on a MaxViT backbone architecture along with data augmentation and transfer learning. While grid attention captures the relationships between regions in the image, block attention captures the contextual information hierarchically, which enables the model to

simultaneously capture fine-grained and large-scale nodule details. The model was validated on IQ-OTH/NCCD dataset with an accuracy of 99.54% and on Kaggle chest CT dataset with an accuracy of 98.41%, with a model size of 18.1 million parameters. The work is a classification study (and not detection) and is not evaluated on LUNA16.

3.1.2 Multi-Scale Transformer for Nodule Diagnosis, MSTD [3]

The work by Zhao et al. [3] presents the novel approach called MSTD, which involves integrating a multi-scale convolutional layer (comprising six parallel layers with various receptive fields) together with a dual branch Transformer fusion layer ("Space Transformer" that attends across spatial patches and "Scale Transformer" that attends across the scale dimension) before classifying detected nodules as either benign or malignant. According to the authors, on the LUNA16 benchmark, the method produces average F1-score 94.5%, specificity 96.5%, and sensitivity 91.1%. It also provides an average improvement of about 1% of AUC, 3.4% of accuracy, and 3.6% of F1-score compared to the previous state of the art solutions. The MSTD algorithm is highly relevant to the proposed architecture regarding Transformer and multi-scale fusion parts, but the model operates only on 2D slices and does not involve any pruning.

3.1.3 Squeeze-and-Excitation Vision Transformer, SE-ViT [4]

According to Xue et al. [4], SE-ViT utilizes a Squeeze-and-Excitation (SE) block prior to a regular Vision Transformer self attention layer such that the image patches (instead of pixels) get weighted based on their informativeness before applying self attention. In LUNA16, SE-ViT achieves 0.863 accuracy, 0.872 precision, 0.876 recall, 0.872 F1-score, and 0.862 AUC. Ablation studies demonstrate that the SE-ViT has better results compared to regular ViT and SE-CNN. Use of Grad-CAM for interpretability and channel attention before self attention mechanism is directly related to the attention module in the current research project; however, SE-ViT is a 2D classification model without any pruning or model compression.

3.2 Lightweight and Memory-Efficient Architectures

3.2.1 MEAA-Net [2]

Qiao and Chuprat [2] introduce the concept of Memory Efficient Asymmetric Attention (MEAA), which involves splitting the feature maps along the batch axis and applying the self attention within each partition individually, in conjunction with gradient checkpointing, for peak reduction in training memory without increase in number of parameters. In their evaluation using the LUNA16 and LIDC-IDRI datasets on five network architectures, their smallest network architecture (61K parameters, 0.76 GFLOPs) attains 80.8% accuracy on the LUNA16 dataset with 11 ms of CPU inference time per single slice, while the MEAA technique improves the peak inference memory on the GPU by 38.1% and the latency by 5.9% compared to the equivalent Transformer version. This paper is very relevant as a direct efficiency comparison as it shows that attention-based lung nodule classification on LUNA16 can be done with orders of magnitude fewer parameters than a ResNet-18 model, which motivates the simultaneous focus of this current project on parameters, FLOPs, memory and latency along with detection accuracy. Unlike the current project, MEAA-Net derives its efficiency from batch-splitting attention and checkpointing instead of weight pruning,

and it is a 2D-slice network.

3.3 Detection-Oriented (YOLO-Family) Architectures

3.3.1 DAFNet: Downsampling Attention Fusion Network [6]

In addition, Li et al. [6] introduces DAFNet, which uses a one-stage detector based on YOLOv11 for the detection of lung nodules. This model consists of two plug-and-play modules, namely, the Adaptive Downsampling Module (ADSM), which utilizes a dual-branch pooling/convolution approach and eliminates the strided convolution approach to retain both low and high-frequency information while reducing the number of parameters, and the Global Attention Module (GAM), which uses channel and spatial attention to enhance the morphological and textural characteristics of nodules. DAFNet claims an accuracy of 93.2% and 90.7% recall rate in self-constructed CT dataset and says that it outperforms the state-of-the-art methods by a wide margin regarding mAP on LUNA16, having an inference speed of 1.5 ms/image and around 2.5 million trainable parameters. Among the presented works, DAFNet is the closest to the real-world application of lightweight detection model with attention mechanism on LUNA16 dataset and the authors clearly mention that the model doesn't have a false-positive reduction or malignancy grading stage, which makes us use the official LUNA16 FROC/CPM detection benchmark for our research.

3.3.2 TiCNet: Transformer in Convolutional Neural Network [7]

TiCNet from Ma et al. [7], which is a transformer-based embedding network placed inside a CNN detection framework to perform pulmonary nodule detection from CT images, shows improved detection accuracy compared to other CNN-only detectors on the LUNA16 dataset. TiCNet is very relevant to our work as a precedence of utilizing Transformer-based contextual modeling together with CNN-based detection backbone.

3.3.3 Swin Transformer for Lung Nodule Detection [8]

Shakuri and Rezvanian [8] conduct an analysis on the feasibility of using a Swin Transformer-based detection pipeline for detecting lung nodules, focusing on the effectiveness of applying hierarchical shifted window self-attention instead of typical CNN backbone networks in the task of detecting nodules. This research is pertinent to the selection of the Transformer architecture to be applied in the step of contextual feature modeling in the proposed architecture.

3.4 A Review of Detection Architectures

3.4.1 YOLO in Healthcare: A Comprehensive Review [5]

Palaniappan et al. [5] conduct a review of peer-reviewed studies on applications of YOLO-based family object detectors in the field of medical image processing, involving fields such as oncology and radiology. This review describes the developments of YOLOv1 to YOLOv8+ (including changes in backbone/necks/head and improved mAP@0.5 from 63.4% in YOLOv1 to 89.2% in YOLOv8), as stated by the review. The authors note that the lightweight versions of YOLO, like YOLOv5s, can process images in real-time frame rates in the field of surgery. Given that this paper is a review and not a methodology, its results cannot be compared on LUNA16 dataset.

The purpose of citing this paper is to provide background on development trends in detection architecture and use of attention/Transformer-enhanced backbones in the field of medical object detection.

4. Problem Statement

Detecting lung nodules using CT images is difficult due to differences in sizes, shapes, densities, and contrast of the nodules, in addition to high class imbalance. Although 2D methods are computationally less expensive, they might lose the inter-slice information, while 3D methods preserve it at much higher computational and memory costs. In existing literature, CNN-Transformer architectures, attention techniques, and lightweight networks have been studied independently but their combination has not been sufficiently investigated within a 2.5D detection pipeline.

The proposed project seeks to develop and test experimentally a framework for detecting lung nodules using 2.5D CNN-Transformer based architecture with the use of attention mechanisms and progressive pruning. The study uses single variable ablation studies conducted on LUNA16 dataset in order to investigate the effect of each element or their combination on the balance between performance (in terms of FROC/CPM, sensitivity and false positives) and efficiency (parameters, FLOPs, memory, latency and sparsity).

5. Objective and Methodology

5.1 Research Objectives

The central research question guiding this project is:

“Can a 2.5D CNN-Transformer lung nodule detection framework, enhanced with attention and progressive pruning, achieve a useful balance between detection performance and computational efficiency on LUNA16?”

5.1.1 Primary Objective

Design and evaluate an end-to-end lung nodule detection model employing a 2.5D CT input representation with CNN–Transformer features extraction.

5.1.2 Secondary Objectives

- Set up an experimental pipeline for establishing a reproducible baseline for lung nodule detection.
- Explore the influence of Transformer-based context modeling over an architecture based solely on CNN.
- Explore the role of attention mechanisms in the detection performance and computational efficiency.
- Design and evaluate a progressive pruning approach.

- Assess the performance of the detector based on LUNA16 metrics (FROC, CPM, sensitivity, FPs per scan).
- Study the computational efficiency (parameters, FLOPs/MACs, memory, latency, sparsity, throughput) of all experimental setups.
- Conduct a series of controlled experiments in order to isolate the effect of each element of the design (CNN baseline; CNN + Transformer; CNN + Transformer + Attention; CNN + Transformer + Attention + Progressive Pruning) and explore its influence on the efficiency/performance balance.

5.2 Proposed Architecture

Figure 5.1 summarizes the proposed architecture.

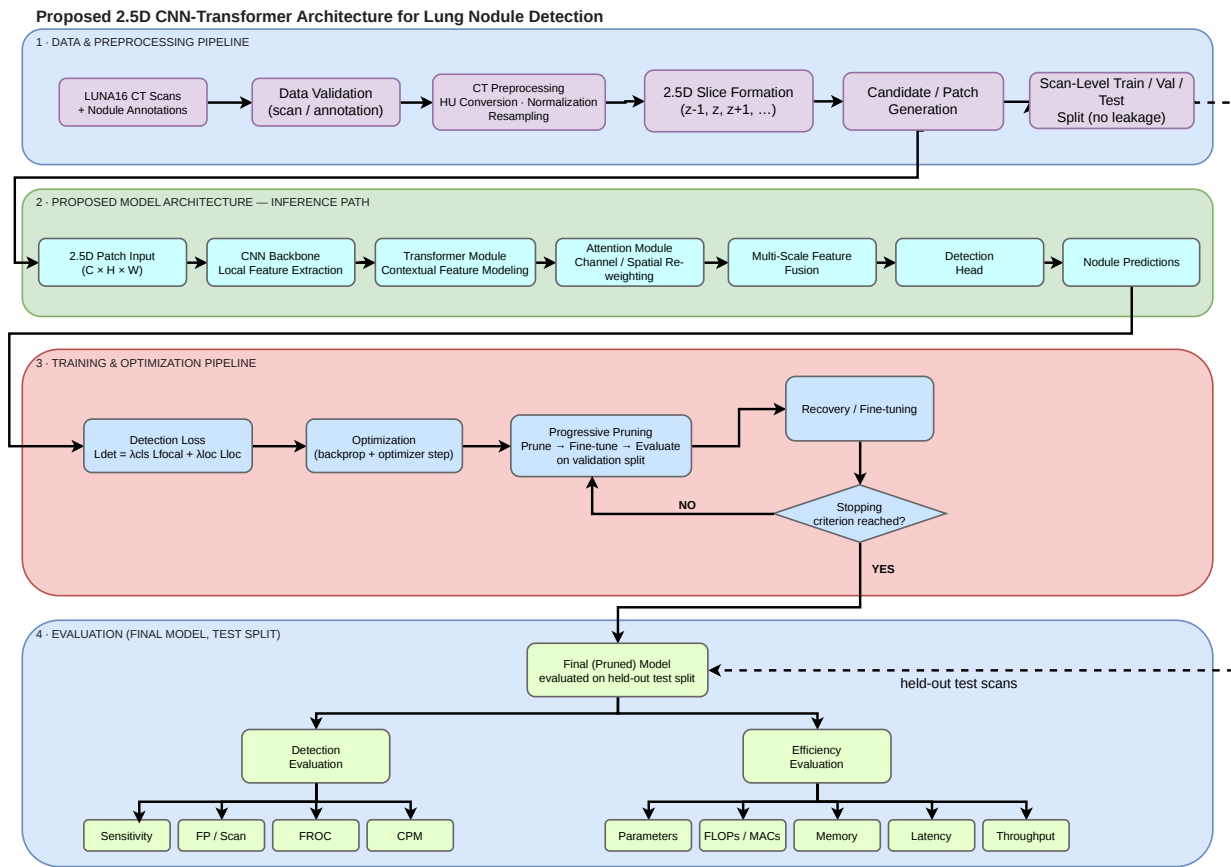


Figure 5.1: Proposed 2.5D CNN–Transformer detection framework.

All the layers are designed to be replaceable individually (data, preprocessing, backbone, attention, fusion, detection head), so that the particular backbone network, depth and embedding of Transformer, attention mechanism used, number of 2.5D slices, and detection head can stay configurable parameters instead of being decided in advance, in line with the architectural constraints of the project. Neither the backbone nor Transformer configuration is considered to be finalized at the moment; the right configuration will be chosen on the basis of results of experiments outlined in Sec. 5.4.

5.3 Proposed Training Objective

The detection objective combines a classification/objectness loss with a localization loss:

$$L_{\text{det}} = \lambda_{\text{cls}} L_{\text{focal}} + \lambda_{\text{loc}} L_{\text{loc}} \quad (5.1)$$

Because the candidate set for nodule detection is heavily dominated by easy negative (non-nodule) samples, a focal loss formulation may be used for the classification/objectness term in Eq. (5.1):

$$L_{\text{focal}} = -\alpha_t (1 - p_t)^\gamma \log(p_t) \quad (5.2)$$

Where

$$p_t = \begin{cases} p, & \text{if } y = 1, \\ 1 - p, & \text{if } y = 0. \end{cases} \quad (5.3)$$

Here, α_t controls class weighting and γ controls the down-weighting of easy examples.

An IoU-based localization loss,

$$L_{\text{loc}} = 1 - \text{IoU}(B, B^*) \quad (5.4)$$

may be used to measure the discrepancy between a predicted box B and the ground-truth box B^* , where

$$\text{IoU}(B, B^*) = \frac{|B \cap B^*|}{|B \cup B^*|} \quad (5.5)$$

is the Intersection over Union between the predicted and ground-truth boxes. The exact localization formulation will be finalized based on the selected detection head.

The overall training objective is the detection objective itself:

$$L_{\text{total}} = L_{\text{det}} \quad (5.6)$$

Substituting Eq. (5.1) into Eq. (5.6) gives:

$$L_{\text{total}} = \lambda_{\text{cls}} L_{\text{focal}} + \lambda_{\text{loc}} L_{\text{loc}} \quad (5.7)$$

The symbols used in Eqs. (5.1)–(5.7) are summarized in Table 5.1.

Symbol	Meaning
L_{total}	Overall training loss
L_{det}	Detection loss
L_{focal}	Focal classification/objectness loss
L_{loc}	Localization loss
λ_{cls}	Weight of the classification/objectness loss
λ_{loc}	Weight of the localization loss
p	Predicted probability of the positive/nodule class
y	Ground-truth class label, $y \in \{0, 1\}$
p_t	Probability assigned to the ground-truth class
α_t	Class-balancing factor
γ	Focal-loss focusing parameter
B	Predicted bounding box
B^*	Ground-truth bounding box
IoU	Intersection over Union

Table 5.1: Notation used in the proposed training objective.

5.4 Progressive Pruning

Instead of the single pruning process at once, the proposed method studies progressive pruning that involves training, evaluation, pruning, and finetuning of the model as demonstrated in Figure 5.2. Once the recovery phase is completed, the model is re-evaluated in terms of its efficiency and detection performance and then the next pruning stage is applied.

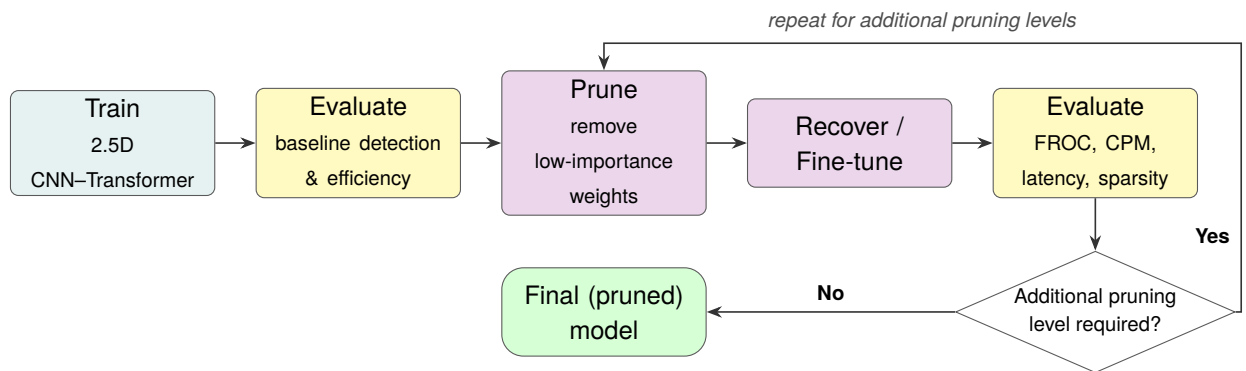


Figure 5.2: Progressive pruning and recovery cycle.

5.5 Evaluation Metrics

However, this is a problem of detection, not classification, so evaluation metrics will be geared towards detection rather than classification performance:

- FROC (Free-response Receiver Operating Characteristic) curve, which plots sensitivity against the number of false positives per image.
- CPM (Competition Performance Metric), the official summary metric for LUNA16, averaged across several false positive rate operating points.
- Efficiency metrics: number of parameters, FLOPs/MACs, model size, peak memory usage, latency, throughput, and sparsity (in the case of pruned networks).

Nothing will be called useful without supporting evidence, and no result will be declared final unless supported by these measurements, regardless of its usefulness.

6. Simulation Platform and Requirements

This section documents the dataset, hardware, and software environment used for implementation and experimentation.

6.1 Dataset

This work utilizes the LUNA16 (LUNG Nodule Analysis 2016) dataset [9] that consists of 888 low-dose chest CT scans from the LIDC-IDRI database, alongside with the corresponding nodule annotation and candidates. The dataset is split on the image level to prevent data leakage between training, validation, and testing sets, and preprocessing parameters (Hounsfield units conversion, normalization, resampling, and lungs region of interest extraction) are saved for each partition separately.

6.2 Hardware Environment

Component	Specification
Machine	MacBook Pro (Apple M3)
Memory	8 GB RAM
Operating System	macOS
Acceleration	Apple MPS backend where supported and stable; CPU execution retained as a fallback for debugging and validation

Table 6.1: Primary development hardware environment.

Because of the 8 GB memory constraint, the implementation is designed to avoid loading the complete LUNA16 dataset into memory, favouring on-demand/lazy loading and memory-conscious

batch sizes; this is treated strictly as an implementation constraint and not as a research contribution in itself.

6.3 Software Environment

Category	Tools / Libraries
Programming language	Python 3.11
Deep learning framework	PyTorch
Numerical / scientific computing	NumPy, SciPy
Data handling	pandas
Machine learning utilities	scikit-learn
Medical image processing	SimpleITK, nibabel
Configuration	PyYAML
Visualization	matplotlib
Testing	pytest
Development environment	VS Code, Conda, Jupyter Notebook
Version control	Git

Table 6.2: Core software stack.

Additional libraries are introduced only when they address a concrete project requirement that cannot reasonably be met with the existing stack, in order to keep the environment reproducible and dependency-light. All experiment configurations (model, training, preprocessing, pruning, and loss settings) are stored as YAML configuration files rather than hard-coded values, so that experiments remain reproducible across runs and machines.

7. Conclusion

In this report, the motivation, background literature, problem statement, and methodology of a 2.5D CNN-Transformer model for detecting lung nodules using the LUNA16 dataset, with an ablation-enabled attention network and a progressive pruning approach, has been described. From the review of literature, it is evident that CNN-Transformer models, attention networks, and lightweight architectures have all separately been studied for classification/detection of lung nodules; however, the use of all three concepts in a 2.5D detection model and a progressively compressed system in conjunction with each other, is relatively understudied.

However, the suggested architecture has been purposefully defined at such a level as to preserve its constituent elements, including CNN structure, Transformer parameters, attention modification, number of 2.5D slices, and pruning policy, as research variables, which should not be predefined. In keeping with the research-integrity principle set for the project, no assertions can be put forward

at this point concerning the impact of the individual elements on performance or efficiency. Instead, all of them will be assessed upon implementation with regard to standard detection metrics (FROC, CPM, sensitivity, FPs per scan) and standard efficiency metrics (parameters, FLOPs, memory, latency, sparsity, throughput).

Similarly, just like all other CAD research works using deep learning techniques, the CAD model proposed in this work is not a clinical diagnostic tool; rather, the predictions made by this model fall under the category of research findings and should in no way be used to override the decision-making ability of radiologists. The next steps for future work in accordance with the phased plan that has been defined for this research work will include the completion of baselines for 2D and 2.5D, implementation and evaluation of the CNN-Transformer model and attention mechanism, validation of progressive pruning technique, and ablation study.

8. References

- [1] B. Ozdemir, E. Aslan, and I. Pacal, “Attention Enhanced InceptionNeXt-Based Hybrid Deep Learning Model for Lung Cancer Detection,” *IEEE Access*, vol. 13, pp. 27050–27063, 2025, doi: 10.1109/ACCESS.2025.3539122.
- [2] L. Qiao and S. Chuprat, “MEAA-Net: Memory-Efficient Asymmetric Attention for Resource-Constrained Lung Nodule Classification,” *IEEE Access*, vol. 14, pp. 18099–18110, 2026, doi: 10.1109/ACCESS.2026.3660056.
- [3] X. Zhao, J. Li, M. Qi, X. Chen, W. Chen, Y. Li, Q. Liu, J. Tang, Z. Han, and C. Zhang, “MSTD: A Multi-Scale Transformer-Based Method to Diagnose Benign and Malignant Lung Nodules,” *IEEE Access*, vol. 13, pp. 16182–16192, 2025, doi: 10.1109/ACCESS.2025.3531001.
- [4] X. Xue, Y. Ma, W. Du, and Y. Peng, “Squeeze-and-Excitation Vision Transformer for Lung Nodule Classification,” *IEEE Access*, vol. 13, pp. 24852–24862, 2025, doi: 10.1109/ACCESS.2025.3529127.
- [5] D. Palaniappan, R. Jain, T. Premavathi, K. Parmar, W. Ghribi, A. M. Ahmed, and N. Ahmad, “YOLO in Healthcare: A Comprehensive Review of Detection Architectures, Domain Applications, and Future Innovations,” *IEEE Access*, vol. 13, pp. 145714–145725, 2025, doi: 10.1109/ACCESS.2025.3599358.
- [6] L. Li, X. Wen, M. Li, X. Wang, X. Feng, and X. Lv, “Downsampling attention fusion network (DAFNet): a You Only Look Once network for lung nodule detection,” *Quantitative Imaging in Medicine and Surgery*, vol. 16, no. 7, p. 532, 2026, doi: 10.21037/qims-2026-1-0050.
- [7] L. Ma, G. Li, X. Feng, Q. Fan, and L. Liu, “TiCNet: Transformer in Convolutional Neural Network for Pulmonary Nodule Detection on CT Images,” *Journal of Imaging Informatics in Medicine*, 2024, doi: 10.1007/s10278-023-00904-y.
- [8] S. Shakuri and A. Rezvanian, “An Efficient Approach to Detecting Lung Nodules Using Swin Transformer,” in *Proc. 19th Iranian Conference on Intelligent Systems (ICIS)*, IEEE,

2024.

- [9] A. A. A. Setio, A. Traverso, T. de Bel, M. S. N. Berens, C. van den Bogaard, P. Cerello, H. Chen, et al., “Validation, comparison, and combination of algorithms for automatic detection of pulmonary nodules in computed tomography images: The LUNA16 challenge,” *Medical Image Analysis*, vol. 42, pp. 1–13, 2017, doi: 10.1016/j.media.2017.06.015.